🧪 Lab: Navigate NGC Catalog and Simulate Pulling a Container

Module 5 – NVIDIA Ecosystem and Tools
Objective: Learn to explore the **NVIDIA NGC Catalog**, identify an AI container (e.g., Triton, PyTorch, TensorRT), and simulate pulling and inspecting it in a local or cloud-based environment.

✅ Learning Goals

- Understand how to browse and search for containers on the NGC catalog
- Identify key metadata: tags, dependencies, CUDA versions, licenses
- Simulate pulling an NGC container using Docker
- Verify container environment and GPU compatibility
- Optional: Run container interactively and inspect setup

🔧 Pre-requisites

- Docker installed on your system
- NVIDIA Container Toolkit (`nvidia-docker`) installed
- A system with GPU access (optional for full testing) or use Google Colab or simulated terminal

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🪜 Step-by-Step Instructions

🔹 Step 1: Visit the NGC Catalog

1. Open your browser and go to:
   🌐 https://catalog.ngc.nvidia.com
2. Browse or search for a container. Try searching for:
   - `"Triton Inference Server"`
   - `"PyTorch"`
   - `"TensorRT"`
3. Click on a container (e.g., **Triton Inference Server**) to open the details page.

> 💡 **Tip**: Every container page includes version tags, supported CUDA versions, usage instructions, and system requirements.

🔹 Step 2: Understand the Container Metadata

1. On the container detail page, note the following:
    - **Container name and publisher**
    - **Supported CUDA version** (e.g., CUDA 12.3)
    - **Latest version tag** (e.g., `nvcr.io/nvidia/tritonserver:24.05-py3`)
    - **Documentation link**
    - **Container pull command**
2. Read the **usage instructions** provided. These often include:
    - How to run the container with volume mounts
    - Sample inference calls
    - Supported model formats (ONNX, TensorFlow, PyTorch, etc.)

🔹 Step 3: Log into NGC (Optional but Recommended)

1. To access private containers or rate-limited ones, create a free NGC account.
   Go to: https://ngc.nvidia.com/signin
2. After creating your account, log in via the terminal:

```
ngc config set
```

    - Enter your **NGC API key** (found under your account settings on the website)
    - Set a name and optional default workspace

🔹 Step 4: Pull a Container Using Docker

1. Open your terminal and simulate pulling the container:
   Example for Triton Inference Server:

```
docker pull nvcr.io/nvidia/tritonserver:24.05-py3
```

2. The image will begin downloading if you have proper permissions and GPU Docker runtime.
3. If you don't have GPU support, you can still simulate pulling (or use `--platform linux/amd64`).

> 🛑 **Troubleshooting Tip**:
> If `nvcr.io` access is restricted, you can use NGC mirrors on Docker Hub like:
> `docker pull nvcr.io/nvidia/pytorch:24.05-py3`

🔹 Step 5: Run the Container and Verify GPU Access

1. Start the container in interactive mode:

```
docker run --rm -it --gpus all nvcr.io/nvidia/tritonserver:24.05-py3 bash
```

2. Inside the container, run:

```
nvidia-smi
```

This should show GPU usage inside the container.

3. (Optional) Check CUDA version:

```
nvcc --version
```

🔷 Step 6: Explore Container Contents (Optional)

- Use `ls`, `cat`, or `vim` to inspect the container's filesystem. Look for:
  - `/opt/nvidia/` or `/opt/tritonserver/`
  - Default model repository
  - Example inference scripts or binaries

🧪 Lab Wrap-Up: What You Should Have Learned

By completing this lab, you should now be able to:

- Navigate the NGC Catalog and interpret container specs
- Simulate or execute a container pull using Docker
- Run the container with GPU access and verify CUDA visibility
- Understand how NVIDIA containers are structured for AI workloads

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